

UChicago REU Notes Week 2

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July 2, 2026

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1 Review: Symmetric Monoidal Categories

Definition 1.1. A category $\mathcal{V}$ is a *monoidal category*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ (called *monoidal product*), and an object $\kappa \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (- \otimes -) \otimes - \xrightarrow{\sim} - \otimes (- \otimes -)$ of functors $\mathcal{V} \times \mathcal{V} \times \mathcal{V} \xrightarrow{\sim} \mathcal{V}$. (So, $(X \otimes Y) \otimes Z \cong X \otimes (Y \otimes Z)$ in a canonical way).
- The *left unit* $\lambda : \kappa \otimes - \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ ($\kappa \otimes X \cong X$ in a canonical way).
- The *right unit* $\rho : - \otimes \kappa \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ ($X \otimes \kappa \cong X$ in a canonical way).

They also satisfy two extra axioms:

- *Triangle Axiom:*

$$\begin{array}{ccc} (X \otimes \kappa) \otimes Y & \xrightarrow{a_{X, \kappa, Y}} & X \otimes (\kappa \otimes Y) \\ & \searrow \rho_X \otimes \text{id}_Y \quad \swarrow \text{id}_X \otimes \lambda_Y & \\ & X \otimes Y & \end{array}$$

- *Pentagon Axiom:*

$$\begin{array}{ccc} ((X \otimes Y) \otimes Z) \otimes W & \xrightarrow{a_{X \otimes Y, Z, W}} & (X \otimes Y) \otimes (Z \otimes W) \\ \downarrow a_{X, Y, Z} \otimes \text{id}_W & & \downarrow a_{X, Y, Z \otimes W} \\ (X \otimes (Y \otimes Z)) \otimes W & & X \otimes (Y \otimes (Z \otimes W)) \\ \searrow a_{X, Y \otimes Z, W} & & \swarrow \text{id}_X \otimes a_{Y, Z, W} \\ & X \otimes ((Y \otimes Z) \otimes W) & \end{array}$$

Based on *MacLane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \rightarrow P_2$, where P_i are (possibly) different orders of performing product on $X_1, \dots, X_n \in \mathcal{V}$. For reference, cf. *Category Theory for Working Mathematicians*.

This guarantees all monoidal category to be equivalent to a *strict monoidal category*, namely all a, λ, ρ above are “identity transformations”, or $1 \otimes X = X, X \otimes 1 = X$, and $(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z)$, so WLOG one can assume the product $\otimes$ is associative.

Definition 1.2. A monoidal category $\mathcal{V}$ is *braided*, if there is a natural isomorphism $B : {}_{-1} \otimes {}_{-2} \xrightarrow{\sim} {}_{-2} \otimes {}_{-1}$ (so $B_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X$ is an isomorphism), together with the *Hexagon axioms* satisfied:

$$\begin{array}{ccc}
(X \otimes Y) \otimes Z & \xrightarrow{a_{X,Y,Z}} & X \otimes (Y \otimes Z) \xrightarrow{B_{X,Y \otimes Z}} (Y \otimes Z) \otimes X \\
\downarrow B_{X,Y} \otimes \text{Id}_Z & & \downarrow a_{Y,Z,X} \\
(Y \otimes X) \otimes Z & \xrightarrow{a_{Y,X,Z}} & Y \otimes (X \otimes Z) \xrightarrow{\text{Id}_Y \otimes B_{X,Z}} Y \otimes (Z \otimes X) \\
\\
X \otimes (Y \otimes Z) & \xleftarrow{a_{X,Y,Z}} & (X \otimes Y) \otimes Z \xrightarrow{B_{X \otimes Y,Z}} Z \otimes (X \otimes Y) \\
\downarrow \text{Id}_X \otimes B_{Y,Z} & & \downarrow a_{Z,X,Y} \\
X \otimes (Z \otimes Y) & \xleftarrow{a_{X,Z,Y}} & (X \otimes Z) \otimes Y \xrightarrow{B_{X,Z} \otimes \text{Id}_Y} (Z \otimes X) \otimes Y
\end{array}$$

$\mathcal{V}$ is a *symmetric monoidal category*, if $B_{Y,X} = B_{X,Y}^{-1}$ for all $X, Y \in \mathcal{V}$.

The two hexagons are uniqueness of 3-cycle permutations on three consecutive objects in a product, hence together with B serving as transpositions, it guarantees unique isomorphism between all permutations of finite products, i.e., it imposes that there is a single way of doing such permutation (so there's no “nontrivial braiding”, when $B_{Y,X} \circ B_{X,Y} : X \otimes Y \rightarrow Y \otimes X \rightarrow X \otimes Y$ is not identity). From my naive understanding of coherence theorem, WLOG we can again treat $X \otimes Y = Y \otimes X$.

Example 1.3. Everyone loves unital commutative ring R , so I use category of R -modules as example :)

Equipped with tensor product $\otimes_R$, and R itself as tensor unit (up to isomorphism), we see $R\text{-Mod}$ forms a symmetric monoidal category due to following natural isomorphisms:

$$\left\{ \begin{array}{ll} (M \otimes_R N) \otimes_R L \cong M \otimes_R (N \otimes_R L) & , \quad (m \otimes n) \otimes \ell \mapsto m \otimes (n \otimes \ell) \\ R \otimes_R M \cong M & , \quad r \otimes m \mapsto rm \\ M \otimes_R R \cong M & , \quad m \otimes r \mapsto rm \\ M \otimes_R N \cong N \otimes_R M & , \quad m \otimes n \mapsto n \otimes m \end{array} \right.$$

Now, observe that $R\text{-Mod}$ is self-enriched, meaning $\text{Hom}_R(-, -) : R\text{-Mod}^{\text{op}} \times R\text{-Mod} \rightarrow R\text{-Mod}$ is a bifunctor (also called an *internal hom functor*). Moreover, there's the tensor-hom adjunction:

$$\text{Hom}_R(M \otimes_R N, L) \cong \text{Hom}_R(M, \text{Hom}_R(N, L)), \quad f \mapsto (m \mapsto f(m \otimes -))$$

This indicates $(- \otimes_R N \dashv \text{Hom}_R(N, -))$ is an adjoint pair, guaranteeing $- \otimes_R N$ commutes with all colimits. For instance, $(M \oplus L) \otimes_R N = (M \otimes_R N) \oplus (L \otimes_R N)$. This gives rise to the following definition:

Definition 1.4. $\mathcal{V}$ is a *symmetric monoidally closed category*, if there exists an *internal hom functor* $\mathcal{V}(-, -) : \mathcal{V}^{\text{op}} \times \mathcal{V} \rightarrow \mathcal{V}$, such that the adjunction $(- \otimes X \dashv \mathcal{V}(X, -))$ holds for all $X \in \mathcal{V}$.

$\mathcal{V}$ is a *symmetric monoidally cocomplete category*, if it's cocomplete, together with $- \otimes X, X \otimes -$ preserves colimits for all $X \in \mathcal{V}$.

If people knows about Yoneda Lemma, welcome to look into the last section, which strengthen the isomorphism in monoidal closeness to natural isomorphism $\mathcal{V}(- \otimes X, -) \cong \mathcal{V}(-, \mathcal{V}(X, -))$ on the bifunctors $\mathcal{V}^{\text{op}} \times \mathcal{V} \rightarrow \mathcal{V}$.

Example 1.5. The category of unital commutative rings $\mathbf{CRing}$ with $\otimes_{\mathbb{Z}}$ as monoidal product is in fact not monoidally closed. Suppose the contrary there exists such bifunctor $C : \mathbf{CRing}^{\text{op}} \times \mathbf{CRing} \rightarrow \mathbf{CRing}$, such that the adjunction holds, then the following isomorphism holds for any rings $R, S \in \mathbf{CRing}$:

$$\text{Hom}_{\mathbf{CRing}}(R, S) \cong \text{Hom}_{\mathbf{CRing}}(\mathbb{Z} \otimes_{\mathbb{Z}} R, S) \cong \text{Hom}_{\mathbf{CRing}}(\mathbb{Z}, C(R, S)) = \{*\}$$

The final implication follows from the fact $\mathbb{Z} \in \mathbf{CRing}$ is initial. Yet, here's a counterexample: $\text{Hom}_{\mathbf{CRing}}(\mathbb{C}, \mathbb{C})$ as at least two maps, including identity ($z \mapsto z$) and conjugation ($z \mapsto \bar{z}$), so we reach a contradiction.

Notice this is distinct from $\mathbf{CRing}$ being monoidally cocomplete: Coequalizer exists in $\mathbf{CRing}$ (by quotienting out the ideal generated by difference of outputs), and $\otimes_{\mathbb{Z}}$ is the coproduct in $\mathbf{CRing}$, hence all colimits exist (cf. *Limit Existence Theorem*). Moreover, colimits commutes with all colimit, in particular so does $\otimes_{\mathbb{Z}}$.

Remark 1.6. For some more abstract examples, we assume $\mathcal{V}$ is monoidally cocomplete. If suitable, also assume internal hom functor exists.

One reason to assume this, is to make sure some of the construction makes sense: In some literature professor May (implicitly) assumes existence of colimits, monoidal product commutes with colimits, and occasionally use internal hom as an example.

Notation 1.7. Let Σ_j denotes the j th symmetric group for all $j \in \mathbb{N}$. In particular, $\Sigma_0 = \Sigma_1 = \{1\}$ as trivial group.

Let Σ denote the category of finite numbers, with objects being $\mathbf{j} := \{1, \dots, j\}$, and homset

$$\text{Hom}_{\Sigma}(\mathbf{i}, \mathbf{j}) = \begin{cases} \emptyset & , \quad i \neq j \\ \Sigma_j & , \quad i = j \end{cases}$$

Also, let $\emptyset \in \mathcal{V}$ be the initial object. Since initial object is a colimit, $\emptyset$ exists, moreover $\emptyset \otimes X \cong \emptyset$ for all $X \in \mathcal{V}$ once assume monoidal cocompleteness.

2 Intro: Operads

As an example, we'll start with category of compactly generated based space $\mathcal{T}^*$, with cartesian product $\times$ as monoidal product, singleton $\{*\}$ as monoidal unit.

Example 2.1. Given $X \in \mathcal{T}^*$, define $\mathcal{E}_X(j) := \text{Hom}(X^j, X)$ for each $j \in \mathbb{N}$.

Let natural numbers $j = j_1 + \dots + j_k$, each $g_i \in \mathcal{E}_X(j_i)$ defines a map $g_i \circ \pi_i : X^j = \prod_{i=1}^k X^{j_i} \rightarrow X$, hence $g_1, \dots, g_k$ induces a product map $\Pi g_i : X^j \rightarrow X^k$. Now, given any $f \in \mathcal{E}_X(k)$, $f \circ g : X^j \rightarrow X^k \rightarrow X$ is an element in $\mathcal{E}_X(j)$. This induces the following structure map:

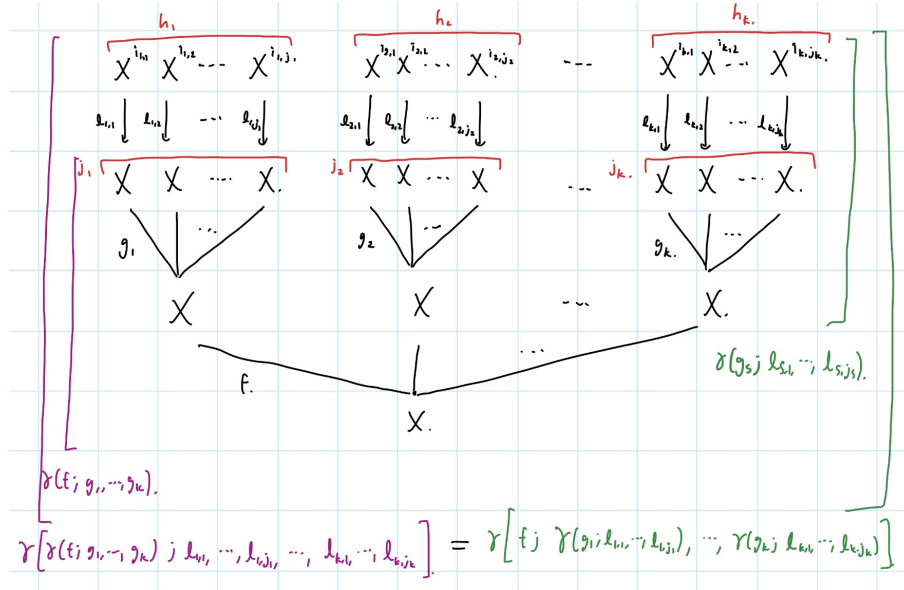
$$\gamma : \mathcal{E}_X(k) \times \prod_{i=1}^k \mathcal{E}_X(j_i) \rightarrow \mathcal{E}_X(j), \quad \gamma(f; g_1, \dots, g_k) := f \circ \Pi g_i$$

There is also a canonical choice of *unit*, namely $\eta : \{*\} \rightarrow \mathcal{E}_X(1)$ by $* \mapsto \text{id}_X$.

Unit here represents an *identity operation*, as one can verify the following two diagram commutes:

$$\begin{array}{ccc} \mathcal{E}_X(j) \times \{*\}^j & \xrightarrow{\sim} & \mathcal{E}_X(j) \\ \downarrow \text{id}_{\mathcal{E}_X(j)} \times \eta^j & \nearrow \gamma & \\ \mathcal{E}_X(j) \times \mathcal{E}_X(1)^j & & \end{array} \quad \begin{array}{ccc} \{*\} \times \mathcal{E}_X(j) & \xrightarrow{\sim} & \mathcal{E}_X(j) \\ \downarrow \eta \times \text{id}_{\mathcal{E}_X(j)} & \nearrow \gamma & \\ \mathcal{E}_X(1) \times \mathcal{E}_X(j) & & \end{array}$$

It also satisfies a giant associativity law, for demonstration purpose I'll draw a tree diagram instead. Given $k, j = j_1 + \dots + j_k$, each j_s associates with $h_s = i_{s,1} + \dots + i_{s,j_s}$, and $i := h_1 + \dots + h_k$. Then, the following combination diagram has no ambiguity, given $f \in \mathcal{E}_X(k)$, $g_s \in \mathcal{E}_X(j_s)$, and $\ell_{s,t} \in \mathcal{E}_X(i_{s,t})$:



Finally, there is a natural right Σ_j -action on $\mathcal{E}_X(j)$ by pre-permuting the input entries, $f \circ \sigma(x_1, \dots, x_j) := f(x_{\sigma^{-1}(1)}, \dots, x_{\sigma^{-1}(j)})$. Which, permutation in $\mathcal{E}_X(k)$ in γ is equivalent to “permuting orders” of blocks $(j_1, \dots, j_k) = \{\{1, 2, \dots, j_1\}, \dots, \{j_1 + \dots + j_{k-1} + 1, \dots, j_1 + \dots + j_{k-1} + j_k\}\}$ in $\{1, \dots, j\}$, and each Σ_{j_s} -action on $\mathcal{E}_X(j_s)$ is equivalent to a permutation only on each block in $(j_1, \dots, j_k)$.

Intuitively, each $\mathcal{E}_X(j)$ encodes all possible way of operating on j points of X (a j -ary operation on X), while each $\mathcal{E}_X(k)$ also encodes operation of combining k (possibly distinct) operations into another operation, in an associative way.

This is called the *Endomorphism Operad of X* . In general, if $\mathcal{V}$ is monoidally closed, the internal hom functor provides a construction of such endomorphism operad in $\mathcal{V}$.

Definition 2.2. Given symmetric monoidal category $\mathcal{V}$, a Σ operad $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{\mathcal{C}(j) \in \mathcal{V}\}_{j \in \mathbb{N}}$, each $\mathcal{C}(j)$ with a right Σ_j -action (a group homomorphism $\Sigma_j^{\text{op}} \rightarrow \text{Aut}_{\mathcal{V}}(\mathcal{C}(j))$), a *unit map* $\eta : 1 \rightarrow \mathcal{C}(1)$, together with composition maps

$$\gamma : \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes \dots \otimes \mathcal{C}(j_k) \rightarrow \mathcal{C}(j_1 + \dots + j_k)$$

that satisfies the following axioms:

- *Unit:*

$$\begin{array}{ccc} \mathcal{C}(j) \otimes \kappa^{\otimes j} & \xrightarrow{\sim} & \mathcal{C}(j) \\ \searrow \text{id}_{\mathcal{C}(j)} \otimes \eta^{\otimes j} & & \nearrow \gamma \\ & \mathcal{C}(j) \otimes \mathcal{C}(1)^{\otimes j} & \end{array} \quad , \quad \begin{array}{ccc} \kappa \otimes \mathcal{C}(j) & \xrightarrow{\sim} & \mathcal{C}(j) \\ \searrow \eta \otimes \text{id}_{\mathcal{C}(j)} & & \nearrow \gamma \\ & \mathcal{C}(1) \otimes \mathcal{C}(j) & \end{array}$$

- *Associativity:* Fix natural numbers $k, j = \sum_{s=1}^k j_s$, define index $g_s := j_1 + \dots + j_s$ for $1 \leq s \leq k$. For each $i_1, \dots, i_{g_k} = i_j$, with sum $i = \sum_{r=1}^j i_r$, and $h_s := i_{g_{(s-1)}+1} + \dots + i_{g_s}$ (so $i = \sum_{s=1}^k h_s$ also), the following diagram commutes:

$$\begin{array}{ccc} \left[\mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(j_s) \right) \right] \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r) \right) & \xrightarrow{\gamma \otimes \text{id}} & \mathcal{C}(j) \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r) \right) \\ \downarrow \text{shuffle} & & \searrow \gamma \\ \mathcal{C}(k) \otimes \left[\bigotimes_{s=1}^k \left(\mathcal{C}(j_s) \otimes \left(\bigotimes_{q=1}^{j_s} \mathcal{C}(i_{g_{(s-1)}+q}) \right) \right) \right] & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes \left(\bigotimes_s \gamma \right)} & \mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(h_s) \right) \\ & & \nearrow \gamma \\ & & \mathcal{C}(i) \end{array}$$

- *Equivariance:*

- Given $\sigma \in \Sigma_k$, $j_1 + \dots + j_k = \Sigma_j$, let $\sigma(j_1, \dots, j_k) \in \Sigma_j$ denotes “permutation of k -blocks of letters”, by $(j_1, \dots, j_k) \mapsto (j_{\sigma^{-1}(1)}, \dots, j_{\sigma^{-1}(k)})$.

Example 2.3. Given $j = 5 = 2 + 3$, the permutation $(1 \ 2) \in \Sigma_2$, $\sigma(2, 3)$ is the permutation $\langle 1, 2; 3, 4, 5 \rangle \mapsto \langle 3, 4, 5; 1, 2 \rangle$ (namely changing the blocks’ order, without changing the order in each block).

Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) & \xrightarrow{\sigma \otimes \text{shuffle}_{\sigma^{-1}}} & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_{\sigma(r)}) \right) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{C}(j) & \xrightarrow{\sigma(j_{\sigma(1)}, \dots, j_{\sigma(k)})} & \mathcal{C}(j) \end{array}$$

- Given $\tau_r \in \Sigma_{j_r}$, denote $\tau_1 \oplus \dots \oplus \tau_k \in \Sigma_j$ by letting each τ_r permuting the j_r th block in j (for instance, let τ_1 acts on $1, \dots, j_1$; let τ_2 acts on $j_1 + 1, \dots, j_1 + j_2$, etc.). Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes (\bigotimes_r \tau_r)} & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{C}(j) & \xrightarrow{\tau_1 \oplus \dots \oplus \tau_k} & \mathcal{C}(j) \end{array}$$

If $\mathcal{C}(0) = \kappa$ (with $\gamma : \mathcal{C}(0) \rightarrow \mathcal{C}$ being $\text{id}_{\mathcal{C}(0)}$), the operad is called *unital operad*. If omit the Σ_j -action, it's called *non- Σ operad*.

Definition 2.4. Let $\mathcal{C}, \mathcal{C}'$ be two operads on symmetric monoidal category $\mathcal{V}$, a *morphism of operads* $\psi : \mathcal{C} \rightarrow \mathcal{C}'$, is a sequence of Σ_j -equivariant morphisms $\psi_j : \mathcal{C}(j) \rightarrow \mathcal{C}'(j)$ in $\mathcal{V}$ (i.e., $\sigma^{-1} \circ \psi_j \circ \sigma = \psi_j$, for all $\sigma \in \Sigma_j$), such that it preserves the operad's structure morphism:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes \dots \otimes \mathcal{C}(j_k) & \xrightarrow{\gamma} & \mathcal{C}(j_1 + \dots + j_k) \\ \psi_k \otimes \psi_{j_1} \otimes \dots \otimes \psi_{j_k} \downarrow & & \downarrow \psi_{j_1 + \dots + j_k} \\ \mathcal{C}'(k) \otimes \mathcal{C}'(j_1) \otimes \dots \otimes \mathcal{C}'(j_k) & \xrightarrow{\gamma'} & \mathcal{C}'(j_1 + \dots + j_k) \end{array}$$

2.1 Categorical Example: Commutative Operad

Let $\mathcal{N}(j) := \kappa$, endows with trivial Σ_j -action. Define $\gamma : \mathcal{N}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{N}(j_r) \right) \rightarrow \mathcal{N}(j)$ to be the canonical isomorphism of unit. This is a simple (yet important definition wise) for defining E_∞ -operad.

2.2 Categorical Example: Associative Operad

Let $\mathcal{M}(j) := \kappa[\Sigma_j] := \coprod_{\psi \in \Sigma_j} \kappa_\psi$, coproduct of $\#\Sigma_j$ copies of κ (each κ_ψ is the unit κ , subscript is for labeling convenience). It has a natural right Σ_j -action, where for each $\sigma \in \Sigma_j$, the identity $\kappa_{\psi\sigma^{-1}} \xrightarrow{\sim} \kappa_\psi$ induces an isomorphism on coproduct $\tilde{\sigma} \in \text{Aut}(\mathcal{M}(j))$:

$$\begin{array}{ccc} \kappa_\psi & \xrightarrow{\sim} & \kappa_{\psi\sigma} \\ \downarrow \iota_\psi & & \downarrow \iota_{\psi\sigma} \\ \coprod_{\psi \in \Sigma_j} \kappa_\psi & \xrightarrow[\exists! \tilde{\sigma}]{} & \coprod_{\psi \in \Sigma_j} \kappa_{\psi\sigma} \end{array}$$

This construction enforces $(\sigma\tilde{\sigma}') = \tilde{\sigma}' \circ \tilde{\sigma}$, which is a group homomorphism $\Sigma_j^{\text{op}} \rightarrow \text{Aut}_{\mathcal{V}}(\mathcal{M}(j))$.

There is a natural unit morphism $\eta : \kappa \rightarrow \kappa[\Sigma_1] = \kappa$ via identity.

Also, each identity morphism $\kappa_\psi \rightarrow \kappa$ induces a trivialization $t : \kappa[\Sigma_j] \rightarrow \kappa$. Combining the action $\psi \in \Sigma_k$ (by identifying concatenation of blocks $(j_{\psi^{-1}(1)}, \dots, j_{\psi^{-1}(k)}) = \{1, \dots, j\}$ in order, also inducing inclusion $\Sigma_{j_s} \hookrightarrow \Sigma_j$), we get the following morphism:

$$\kappa_\psi \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}] \right) \xrightarrow{\sim} \bigotimes_{s=1}^k \kappa[\Sigma_{j_{\psi^{-1}(s)}}] \xrightarrow{t^{k-1}} \kappa[\Sigma_{j_{\psi^{-1}(s)}}] \rightarrow \kappa[\Sigma_j]$$

Running through all $\psi \in \Sigma_k$, this induces a morphism $\gamma : \coprod_{\psi \in \Sigma_k} \left[\kappa_\psi \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}] \right) \right] \rightarrow \kappa[\Sigma_j]$, and use the fact $\otimes$ commutes with colimits, this is equivalent to a morphism $\gamma : \kappa[\Sigma_k] \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}] \right) \rightarrow \kappa[\Sigma_j]$.

Verify $\mathcal{M}$ forms an operad is a long tedious work, unless many people request the proof we'll omit it here :)

As a side note, this is also crucial for defining A_∞ -operad.

2.3 Topological Example: Stasheff Associahedra (To do)

Given a based space X , its loop space $Y = \Omega X := F(S^1, X)$ has a product via concatenation: $Y^2 \rightarrow Y$ by $(g, f) \mapsto g \cdot f$, where

$$(g \cdot f)(t) = \begin{cases} f(2t) & 0 \leq t \leq \frac{1}{2} \\ g(2t - 1) & \frac{1}{2} < t \leq 1 \end{cases}$$

when viewing $S^1 = I/\partial I$.

3 Operad Algebra

As a motivating example, let's start with the endomorphism operad:

Example 3.1. Take $X \in \mathcal{T}^*$ and its endomorphism operad, there is a collection of canonical maps $\theta : \mathcal{E}_X(j) \times X^j \rightarrow X$, given by $\theta(f; x_1, \dots, x_j) := f(x_1, \dots, x_j)$.

It inherits the associativity, because the resulting action on X^j disregards if we plugin the entries first then combine the multimorphisms, or combine the multimorphisms then plugin the entries.

The identity action reflects its name, as $\{*\} \times X \rightarrow \mathcal{E}_X(1) \times X \rightarrow X$ by $(*; x) \mapsto (id_X; x) \mapsto id_X(x) = x$ is an isomorphism.

Also, notice each X^j has a canonical left Σ_j -action by permutation, $\sigma(x_1, \dots, x_j) = (x_{\sigma(1)}, \dots, x_{\sigma(j)})$, and notice that the Σ_j -action on $\mathcal{E}_X(j) \times X^j$ by $(f\sigma; (x_1, \dots, x_j)) = (f; \sigma^{-1}(x_1, \dots, x_j))$, so the Σ_j -action can be communicated between the product's entries.

Definition 3.2. Given an operad $\mathcal{C}$ over a symmetric monoidal category $\mathcal{V}$, a $\mathcal{C}$ -algebra is an object $A \in \mathcal{V}$, with collections of morphisms $\theta : \mathcal{C}(j) \otimes A^j \rightarrow A$, that satisfies the following axioms:

- *Associativity:* Given $j = \sum_{i=1}^k j_i$, the following diagram commutes:

$$\begin{array}{ccc}
 \left[C(k) \otimes \left(\bigotimes_{i=1}^k C(j_i) \right) \right] \otimes A^{\otimes j} & \xrightarrow{\gamma \otimes id_{A^{\otimes j}}} & C(j) \otimes A^{\otimes j} \\
 \downarrow \text{shuffle} & & \searrow \theta \\
 C(k) \otimes \left[\bigotimes_{i=1}^k (C(j_i) \otimes A^{\otimes j_i}) \right] & \xrightarrow{id_{C(k)} \otimes \theta^k} & C(k) \otimes A^{\otimes k} \\
 & & \nearrow \theta \\
 & & A
 \end{array}$$

(Note: This part uses the isomorphism $A^{\otimes j} \cong \bigotimes_{i=1}^k A^{\otimes j_i}$).

- *Unit:* The following diagram commutes:

$$\begin{array}{ccc}
 \kappa \otimes A & \xrightarrow{\sim} & A \\
 \searrow \eta \otimes id_A & & \nearrow \theta \\
 & C(1) \otimes A &
 \end{array}$$

- *Equivariance:* Given any permutation $\sigma \in \Sigma_j$, the following diagram commutes:

$$\begin{array}{ccc}
 C(j) \otimes A^{\otimes j} & \xrightarrow{\sigma \otimes \text{shuffle}_\sigma} & C(j) \otimes A^{\otimes j} \\
 \searrow \theta & & \swarrow \theta \\
 & A &
 \end{array}$$

For instance, any loop space is automatically a $\mathcal{K}$ -space, acted by the Stasheff Associahedras.

Example 3.3. A $\mathcal{N}$ -algebra $A \in \mathcal{V}$ is a *commutative monoid* in the symmetric monoidal category: It has a product morphism $\theta : \mathcal{N}(2) \otimes A^2 \cong A \otimes A \rightarrow A$, and a unit morphism $\theta : \kappa = \mathcal{N}(0) \rightarrow A$.

- It's associative based on the following diagram and something analogous:

$$\begin{array}{ccc}
\mathcal{N}(2) \otimes \mathcal{N}(1) \otimes \mathcal{N}(2) \otimes A^3 & \xrightarrow{\gamma \otimes \text{id}_{A^3}} & \mathcal{N}(3) \otimes A^3 \\
\downarrow \text{shuffle} & & \searrow \theta \\
\mathcal{N}(2) \otimes (\mathcal{N}(1) \otimes A) \otimes (\mathcal{N}(2) \otimes A^2) & \xrightarrow{\text{id}_{\mathcal{N}(2)} \otimes \theta^2} & \mathcal{N}(2) \otimes A^2 \\
& & \nearrow \theta \\
& & A
\end{array}$$

This diagram is formally saying the 3-ary multiplication is equivalent to “multiply” the last two entries, then multiply with the first entry. Interchange the 1 and 2, we get the other order of multiplication.

- It's commutative because taking transposition $\tau \in \Sigma_2$ (with trivial action on $\mathcal{N}(2)$), the equivariance diagram provides:

$$\begin{array}{ccc}
\mathcal{N}(2) \otimes (A_{(1)} \otimes A_{(2)}) & \xrightarrow{\tau \otimes \text{shuffle}_\tau} & \mathcal{N}(2) \otimes (A_{(2)} \otimes A_{(1)}) \\
& \searrow \theta & \swarrow \theta \\
& A &
\end{array}$$

- Unit morphism by definition satisfies the following:

$$\begin{array}{ccccc}
A \cong \mathcal{N}(0) \otimes A & \xrightarrow{\theta \otimes \text{id}_A} & A \otimes A \cong \mathcal{N}(2) \otimes A^2 & \xleftarrow{\text{id}_A \otimes \theta} & A \cong A \otimes \mathcal{N}(0) \\
& & \downarrow \theta & & \\
& & A & &
\end{array}$$

Example 3.4. A $\mathcal{M}$ -algebra $A \in \mathcal{V}$ is a *monoid* in the symmetric monoidal category. Unit morphism follows as $\theta : \mathcal{M}(0) = \kappa \rightarrow A$, and multiplication morphism is $\mu : \kappa_1 \otimes A^2 \rightarrow \kappa[\Sigma_2] \otimes A^2 \xrightarrow{\theta} A$. Associativity and unity are similar to above, while this is no longer commutative in general, because there is a nontrivial action of Σ_2 on $\kappa[\Sigma_2]$.